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January 29, 1986
SUPERSEDING
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July 16, 1981

FEDERAL SPECIFICATION

COMPASSES, MAGNETIC: SURVEYOR'S AND TRANSIT, POCKET; WITH OPTIONAL BALL AND SOCKET JOINT OR BALL AND SOCKET HEAD, AND JACOB'S STAFF

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of magnetic compasses, and includes the optional ball and socket joint, or ball and socket head, and Jacob's staff.

1.2 Classification. Magnetic compasses covered by this specification shall be the following types, classes, and styles as specified (see 6.2):

Type I - Surveyor's magnetic compass using ball and socket joint.
Type II - Magnetic compass (pocket transit) using ball and socket head.
Class 1 - 0 to 90 degree quadrant dial.
Class 2 - 0 to 360 degree circle dial.
Style A - With ball and socket joint, or ball and socket head, and Jacob's staff.
Style B - Without ball and socket joint, or ball and socket head, and without Jacob's staff.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

KK-L-154	- Leather, Cattle hide (Bag and Case).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Federal Standards:

FED-STD-H28	- Screw-Thread Standards For Federal Services.
FED-STD-123	- Marking for Shipment (Civil Agencies).

AMSC N/A FSC 6675

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-V-3144	- Vials, Level.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-889	- Dissimilar Metals.

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Federal Regulations:

41 CFR 1-1.25 - Recovered Materials.

(The Code of Federal Regulations [CFR] is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

American Society for Testing and Materials (ASTM):

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Description. The compasses shall consist of a compass box, clinometer, magnetic needle, circular level vial, mechanism to permit declination adjustment, front and rear sights, graduated circles and scales, and when specified herein, a Jacob's staff (see 3.8), ball and socket joint, or ball and socket head (see 3.9) and a carrying case (see 3.10).

3.2 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Metals. Metals, except that of the needle, shall be nonferrous and corrosion resistant. Needles shall be plated or otherwise durably coated to render them corrosion resistant.

3.3 Threads. When specified herein, screw threads shall be Unified National Coarse (UNC) or Unified National Fine (UNF) in accordance with FED-STD-H28. All internal threads shall be countersunk a distance of not less than one-half the pitch of the thread at an angle of 45 degrees, plus or minus 5 degrees, with the axis of the thread. All external threads shall be chamfered at an angle of not less than 30 degrees and not more than 45 degrees with the axis. Countersinking and chamfering shall be concentric with the threads.

3.4. Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 (41 CFR 1-1.25) to the maximum extent practical.

3.4.1 Material deterioration and control. The compasses shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

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3.4.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-C-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.5 Compass box. The compass box shall be metal and shall have a transparent window affording full visibility of the azimuth scale, the clinometer level vial scale and index, and the compass level vial. The compass box shall have an external screw driver-type adjustment so that the azimuth scale can be rotated with respect to the index.

3.6 Magnetic needle. The magnetic needle shall have magnets permanently attached or shall be magnetized to be sensitive to the earth's magnetic field. The needle shall be designed and constructed to assure long retention of magnetism and shall indicate magnetic North when the box is rotated 360 degrees. The needle shall be mounted with a setting or "jewel" polished smooth inside and without cracks. The needle shall be freely supported on a pivot of nonmagnetic hardened steel that provides minimum friction. The needle shall be designed in such a manner as to permit use or be provided with a balance weight which can be moved readily to any position and on either end to assure correct balance in any latitude between 85 degrees North magnetic latitude and 65 degrees South magnetic latitude. The needle shall be self-aligning to accurately indicate direction of magnetic North and shall be marked to readily indicate North. The ends of the needle shall be tapered to a fine point or marked so that 1/2 degrees can be estimated. There shall be no apparent difference in the reading of the ends of the needle. The needle shall be equipped with a lift mechanism to secure it when not in use. The damping time of the needle shall not exceed 30 seconds.

3.7 Dial graduations.

3.7.1 Class 1 dial. The class 1 compass shall be graduated in quadrants, the zero graduation shall be North and South and shall increase 90 degrees in both directions to East and West.

3.7.2 Class 2 dial. The class 2 compass dial shall be graduated in a complete circle. The North point shall be zero and the graduations shall increase in a counterclockwise direction to 360 degrees.

3.8 Jacob's staff. The Jacob's staff shall be a hardwood or nonferrous metal staff, not less than 54 inches long, threaded 5/8-11 UNC-2A for a type I compass and 1/4-20 UNC-2A for type II compass to mate with the female end of the ball and socket joint or ball and socket head as specified in 3.9. The other end of the

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staff shall be equipped with a pointed metal shoe for penetrating the ground. The staff shall be one piece or collapsible and shall weigh not more than 4 pounds.

3.9 Ball and socket joint or ball and socket head. A ball and socket joint or ball and socket head to attach the compass to the Jacob's staff shall be constructed to permit tilting the compass through 90 degrees from the horizontal to the vertical plane and also permit full rotation of the compass in azimuth through 360 degrees. The ball joint shall be equipped with a knurled collar to tighten the joint in tilted position and a knurled thumbscrew to tighten the joint in azimuth position. The male end of the ball joint shall be threaded 1/2-20 UNF-2A to mate with that in the compass base female recessed fitting, which shall be at least 1/4-inch deep for a type I compass and the type II compass shall be furnished with a bracket to attach the compass to the ball joint. The other end of the ball joint shall be threaded 5/8-11 UNC-2B for a type I compass and 1/4-20 UNC-2A for type II compass to fit the Jacob's staff specified in 3.8.

3.10 Carrying case. When specified (see 6.2), a sewed leather case with a sling strap shall be provided to hold the compass. When applicable, the carrying case shall be of leather conforming to KK-L-154, type II, class I, style A, form 1, 1/16-inch thick. A holdown strap and a nonferrous metal fastening stud shall be provided for the closing flap. All edges of the leather shall be stained and creased. All stitched edges shall be burnished and varnished.

3.11 Identification and marking. The compass shall be marked for identification in accordance with MIL-STD-130.

3.12 Treatment and painting. The portions of the compass, ball and socket joint, or ball and socket head, and Jacob's staff normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, color at contractor's option.

3.13 Type I, surveyor's magnetic compass.

3.13.1 Beveled knurled revolving ring. The beveled knurled revolving ring shall be aluminum and shall be graduated in single degrees both ways from zero in four 90-degrees quadrants, with a sighting mark at each quadrant.

3.13.2 Compass dial. The compass dial shall be marked to indicate the four cardinal compass points. East and west points shall be transposed 180 degrees from normal map positions. A clinometer scale for reading vertical angles shall be graduated in 1-degree increments for 180 degrees, and numbered every 10 degrees in both directions from zero to 90 degrees. The zero point shall coincide with the west cardinal point on the compass dial.

3.13.3 Raised compass ring. The raised compass ring shall be graduated in single degrees and numbered every 10 degrees in both directions from zero to four 90-degree quadrants. A mechanism shall be provided to adjust the azimuth of the raised compass reading for declination. The mechanism shall be enclosed within the compass box except for the operating pinion, which shall extend through to

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the outside of the compass box terminating in a slotted screwhead to facilitate the adjustment from outside the compass box. The mechanism shall be nonmagnetic, corrosion-resistant metal and shall be constructed in such a manner that dust or moisture cannot enter the compass box where the adjustment screw terminated. A fixed index pointer shall be permanently secured to the inside wall of the compass box and shall extend over the raised compass ring to serve as a reference point for adjusting the raised compass ring. The index pointer shall coincide with the north point of the compass dial.

3.13.4 Tolerances. The tolerances for the beveled knurled revolving ring, compass dial, and raised compass ring shall be plus or minus 10 minutes per degree graduation, with no more than plus or minus 30 minutes per quadrant. The error shall be noncumulative over the entire circle.

3.13.5 Pendulum clinometer. A clinometer pendulum shall be enclosed in the compass box for reading vertical angles when the compass is placed vertically on its edge on a level surface. The pendulum shall swing freely from an axis concentric with the pivot of the compass needle. The vertical angle shall be indicated by the position of the pendulum against the clinometer scale on the compass dial.

3.13.6 Spirit level. A circular hermetically sealed spirit level for leveling the compass shall be located and secured to the inside of the compass box. The level shall not cover any essential inscriptions and shall conform to MIL-V-3144.

3.13.7 Compass box. The compass box shall not be more than 4 inches in diameter.

3.13.8 Magnetic needle. The magnetic needle shall be not less than 3 inches long and shall be equipped with a stop and a jewel setting.

3.13.9 Folding sights. Two folding sights shall be secured to the compass box by not less than two screws for each sight. The sights shall be hard forged brass or bronze and shall be constructed to permit their being raised to a vertical position for sighting objects or to be folded down flat over the compass glass when not in use. When in the vertical sighting position, the sights shall project above the revolving ring on top of the compass box to a distance equal to the diameter of the revolving ring. The center of the sights shall be positioned in the north-south axis of the compass dial and also to the centerline of the base parallel to two of the beveled edges of the base. The rear (south) folding sight shall have a narrow vertical peep slot centered in the sight. The length of the slot shall be 75 percent of the total length of the sight. The rear sight shall be located on the south end of the compass box and in line with the north-south axis. The front (north) sight shall have a wider slot, 50 percent of the width of the sight, equipped with a vertical sighting line from 0.010 to 0.015 inch wide running the length of the slot. The line and slot shall be centered on the north-south axis of the compass dial at the north end of the compass box.

3.13.10 Compass base. A 4-1/2- by 4-1/2-inch aluminum or nickel silver base shall be secured to the compass box by not less than four countersunk screws with

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the compass box positioned in the center of the base. The relative position of the base and compass box, in azimuth, shall be that the flat-sides of the base shall be parallel to the north-south and east-west axes of the compass dial. The base shall be smooth and flat on the bottom to permit close and accurate positioning on a flat surface. The base shall have cut away or recessed space for screw adjustment devices and fittings. An internal fitting threaded 1/2-20 UNF-2B, shall be provided in the base to accommodate the male end of the ball and socket joint or ball and sockethead specified in 3.9. The fitting shall not permit dust or moisture to enter the compass box. The base shall have four beveled and graduated edges. Two adjacent edges shall be graduated in 1/2 degree increments, plus or minus 30 seconds, through 90 degrees and numbered every 10 degrees up to 80 degrees in both directions. The 45-degree mark shall be at the intersecting corner of the two adjacent edges. Another edge shall be graduated in 1/8-inch increments, plus or minus 1/64-inch and numbered every 1 inch. The other edge shall be graduated in 1/10-inch increments, plus or minus 1/100-inch, and numbered every 1 inch.

3.14 Type II, magnetic compass (pocket transit).

3.14.1 Compass dial. The compass dials shall be graduated in 1-degree increments and numbered every graduated 10 degrees. Tolerance shall be plus or minus 10 minutes per degree graduation, with no more than plus or minus 30 minutes per quadrant. The error shall be noncumulative over the entire circle.

3.14.2 Declination of adjustment. A mechanism shall be provided to adjust the position of the azimuth of the compass dial not less than 50 degrees in either direction.

3.14.3 Index pointer. An index pointer shall be permanently secured to the inside wall of the compass box and shall extend over the compass dial. The index pointer shall be in line with the front sight and shall coincide with the north point of the compass.

3.14.4 Clinometer. The clinometer shall consist of a tubular vial, a clinometer scale and a vernier index mounted on the compass box in such a manner that the clinometer may be rotated by an external leveler. The clinometer shall have its zero at the West cardinal point. Graduations shall extend on both sides of the zero and shall be in degree and percent declination. The degree scale shall be graduated every 1 degree, plus or minus 1/2 degree, shall extend 90 degrees on each side of zero, and shall be numbered every 10 degrees. The vernier shall permit reading to 5 minutes. The percent scale shall extend 100 percent in both directions. Graduations shall be by 5 percent increments, except that gradations may be by 10 percent increments from 60 to 100 percent. Each 20 percent graduation shall be numbered.

3.14.5 Spirit level. A circular hermetically sealed spirit level shall be secured to the inside of the compass box in the Southeast quadrant. The level shall conform to MIL-V-3144.

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3.14.6 Dimensions and weights. The closed compass shall be not more than 4 by 4 inches, with rounded corners, by not more than 1-3/8 inches thick. The compass shall weigh not more than 10 ounces.

3.14.7 Magnetic needle. The magnetic needle shall be between 2-1/8 and 2-1/4 inches long, with jeweled center and a stop.

3.14.8 Mirror and cover. A mirror shall be bonded or otherwise mounted on a hinged cover attached to the south end of the compass box. The cover and the mirror shall provide a 3/8- by 15/32-inch clear glass window, an engraved center-line which will serve as an alternate front sight, and a folding front sight tip. Hinge fringe shall be such as to permit setting of the mirror at any angle between full-closed and full-open position of the cover. Hinge stops shall be provided at 180 degrees open position. In the closed position, the cover shall actuate a sealed plunger and lever mechanism in the compass box to lift the magnetic needle from its pivot and hold the needle in place against the compass box window.

3.14.9 Cover latch. The cover shall latch into the closed position by use of a spring detent or closely controlled mechanical interference. The cover shall open by use of fingergrips (either natural or machined in) or by use of a push-to-open plunger-type spring release. When no mechanical opening aid is provided, fingergrips will be provided on the exterior of the cover and compass box.

3.14.10 North sight. The sight attached to the north end of the compass box shall fold through 180 degrees plus 10 degrees with a tip that folds 55 degrees plus or minus 7 degrees upward from open position. The tip shall be 3/8-inch long plus or minus 1/32-inch. The rear sight shall be 2-3/4-inch long plus or minus 1/32-inch, including the folding tip.

3.15 Workmanship. The magnetic compasses shall have a uniform unmarred finish. Each component shall be constructed to facilitate an accurate fit of mating parts. Internal and external parts shall be free from residue of fabrication and shall be free of any defects which affect serviceability or appearance.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence

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of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level I.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.3 Test. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.1. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The compasses shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Recovered materials not in accordance with Public Law 94-580 (41 CFR 1-1.25).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes, or treatment.
106. Screw threads not as specified.
107. Compass box not as specified.
108. Dial graduations not as specified.
109. Jacob's staff not as specified.
110. Ball and socket joint or ball and socket head not as specified.
111. Beveled knurled ring not as specified.
112. Compass dial not as specified.
113. Raised compass ring not as specified.

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114. Tolerances not as specified.
115. Pendulum clinometer not as specified.
116. Spirit level not as specified.
117. Magnetic needle not as specified.
118. Folding sights not as specified.
119. Compass base not as specified.
120. Carrying case for type I or type II not as specified.
121. Declination adjustment mechanism not as specified.
122. Index pointer not as specified.
123. Clinometer not as specified.
124. Mirror and cover not as specified.
125. Cover latch not as specified.
126. North sight not as specified.
127. Dimensions and weight not as specified.
128. Ball and socket head for type II compass not as specified.
129. Workmanship not as specified.

Minor

201. Identification marking not as specified.
202. Treatment and painting not as specified.

4.4.2 Tests.

4.4.2.1 Magnetic needle. The compass shall be placed in a level position with the needle stop released to allow the needle to rotate freely and to settle in position indicating the North magnetic pole. The compass shall then be rotated 360 degrees. Inability of the magnetic needle to maintain its position pointing to magnetic North through the complete rotation shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

130. Preservation and packaging not as specified for level A and B.
131. Consolidation not as specified for level A or B.
132. Contents not blocked or cushioned as specified for level A or B.
133. Boxes not closed in accordance with specified method for level A or B.
134. Preservation and packaging and packing not in accordance with the referenced document as specified for commercial.

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135. Compasses of unlike description packed together for level A, B, or commercial.
136. Marking missing, illegible, incorrect, or incomplete for level A, level B, or commercial.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I. Each type I compass, contained in the carrying case, when furnished (see 3.10 and 6.2); ball and socket joint, or ball and socket head, if furnished (see 3.9), also contained in the carrying case; and the technical publications, when provided (see 6.3), shall be preserved in accordance with MIL-P-116, method IA-8.

5.1.1.1.1 Jacob's staff. The Jacob's staff, when furnished (see 3.8), shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.1.2 Type II. Each type II compass, contained in the carrying case when furnished (see 3.10 and 6.2), and the technical publications when provided (see 6.3), shall be preserved together in accordance with MIL-P-116, method IA-8.

5.1.1.3 Consolidation. Type I compasses, furnished with a ball and socket joint or ball and socket head and a Jacob's staff, shall be consolidated together as a complete transit in a box conforming to PPP-B-636, type CF, class weather-resistant, grade W6c. Type II compasses of like description furnished without a ball and socket joint or ball and socket head and a Jacob's staff shall be consolidated together in quantities which do not exceed 30 pounds. The box shall conform to PPP-B-636, grade W6c. Contents shall be blocked and cushioned within the box so as to prevent movement. The box shall be closed in accordance with method V of the appendix to the box specification.

5.1.2 Level B.

5.1.2.1 Military activities. Types I and II compasses shall be preserved as specified in 5.1.1, except that method IC-1 and IC-3 shall be used instead of method IA-8. Jacob's staffs shall be preserved as specified for level A.

5.1.2.2 Civil agencies. Each compass with carrying case and accessories, when applicable, shall be packaged in a close fitting water-proof or grease-proof, water-proof bag. The bag shall be sealed in a manner that will provide strength equal to the bag material. The Jacob's staff, when furnished, shall be packaged as specified herein for the compass with carrying case. When the compasses are furnished with a ball and socket joint or ball and socket head to form a complete transit, one complete set of packaged components shall be consolidated in a close fitting box conforming to PPP-B-636, class domestic, style optional. Blocking and bracing shall be utilized as necessary to prevent movement or damage of the box contents during shipment and storage.

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5.1.3 commercial. Compasses and Jacob's staffs shall be preserved and packaged in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Compasses of like description, with accessories as applicable, preserved and packaged as specified in 5.1, either individually or consolidated, shall be packed in close fitting boxes conforming to PPP-B-601, overseas type, style optional. Gross weight of boxes shall not exceed 200 pounds. Boxes shall be closed and strapped in accordance with the appendix to the box specification except that strapping shall be flat steel, finish B.

5.2.2 Level B.

5.2.2.1 Military activities. Compasses of like description, with accessories as applicable, preserved and packaged as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, class weather resistant, style and grade optional. Weight of boxes shall not exceed the limitation specified for the category of boxes. Box closure and sealing shall be in accordance with the appendix to the box specification.

5.2.2.2 Civil agencies. Compasses of like description, with accessories as applicable, preserved and packaged as specified in 5.1, shall be packed as specified in 5.2.2.1, except the box shall be class domestic.

5.2.3 Commercial. The compasses of like description with accessories as applicable preserved and packaged as specified in 5.1, shall be packed in containers in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Civil agencies. Marking shall be in accordance with FED-STD-123.

5.3.2 Military activities. Marking for levels A and B shall be in accordance with MIL-STD-129. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be placed on shipping container.

6. NOTES

6.1 Intended use. Type I compasses are intended for use in topographical surveying field work done by the US Forest Service. Type II compasses are used in geology and survey reconnaissance.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

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a. Title, number and date of this specification.
b. Type, class and style required (see 1.2).
c. When a carrying case is required (see 3.10).
d. Degree of preservation and packaging and degree of packing required (see 5.1 and 5.2).

6.3 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each compass.

6.4 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTEREST:

Custodians

Army - ME
Air Force - 99

Review Activities

Air Force - 82
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
USDA-AFS, CSC
INT-BLM

PREPARING ACTIVITY:
Army - ME

Project 6675-0750

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